30/09/2026

Code-C Phase-3

Aakash

Medical|IIT-JEE|Foundations

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NCERT Booster Test Series for NEET-2027 (XII-Passed)_NBTS01C

MM : 720

Time : 180 Min.

Topics Covered:

Physics: Units & Measurements, Motion in a Straight Line, Motion in a Plane, Laws of Motion, Work, Energy & Power, System of Particles & Rotational Motion

Chemistry: Some Basic Concepts of Chemistry, Structure of Atom, Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure, Thermodynamics

Botany: Cell: The Unit of Life, Cell Cycle & Cell Division, The Living World, Biological Classification, Morphology of Flowering Plants

Zoology: Structural Organisation in Animals – Animal Tissues, Biomolecules, Breathing & Exchange of Gases, Body Fluids & Circulation

General Instructions :

Duration of Test is 3 hrs.

The Test consists of 180 questions. The maximum marks are 720.

There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.

Each question carries 4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.

Use blue/black ballpoint pen only to darken the appropriate circle.

Mark should be dark and completely fill the circle.

Dark only one circle for each entry.

Dark the circle in the space provided only.

Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

1. A substance of mass 42.60 g occupies 1.2 cm^3 of volume. The density of substance (in g cm^{-3}) with correct number of significant figures is

(1) 36
(2) 35.5
(3) 35.0
(4) 35

2. The measured values of two resistance R_1 and R_2 are as $R_1 = 100 \pm 0.3\ \Omega$ and $R_2 = 50 \pm 0.2\ \Omega$. Then match column I with column II

Column I		Column II	
(A)	Max. percentage error in R_1	(P)	1%
(B)	Max. percentage error in R_2	(Q)	0.33%
(C)	Max. percentage error in $R_1 + R_2$	(R)	0.4%
(D)	Max. percentage error in $R_1 - R_2$	(S)	0.3%

(1) A(P), B(Q), C(R), D(S)
(2) A(S), B(R), C(P), D(Q)
(3) A(S), B(R), C(Q), D(P)
(4) A(S), B(P), C(Q), D(R)

3. The only quantity which has negative dimension of mass is

(1) Angular momentum
(2) Electrical conductivity
(3) Gravitational constant
(4) Both (2) and (3)

4. In a new system of units, 5 kg is unit of mass, 10 m is unit of length and 5 second is unit of time. The value of 2 pascal (N m^{-2}) in this new system is

(1) 100 units
(2) 50 units
(3) 25 units
(4) 4 units

5. If velocity (*V*), mass (*M*) and time (*T*) are taken as fundamental quantities, then the dimensional formula of gravitational potential energy would be

(1) $[MV^2T^0]$
(2) $[MV^{-2}T^1]$
(3) $[M^2VT^0]$
(4) $[M^{-2}VT^0]$

6. Two quantities *A* and *B* have measured values as $A = a \pm \Delta a$ and $B = b \pm \Delta b$. If *Z* is represented as $Z = A - B$, then

(1) $\frac{\Delta Z}{Z} = \frac{\Delta a}{a} + \frac{\Delta b}{b}$
(2) $\Delta Z = \Delta a - \Delta b$
(3) $\frac{\Delta Z}{Z} = \frac{\Delta a + \Delta b}{a + b}$
(4) $\Delta Z = \Delta a + \Delta b$

7. If force (*F*), length (*L*) and time (*T*) are assumed to be fundamental quantities then dimensional formula for mass is

(1) $[FL^{-1}T^2]$
(2) $[F^{-1}L^2T^{-2}]$
(3) $[F^{-1}L^2T^2]$
(4) $[FL^1T^{-2}]$

8. A 200 m long train moving with velocity 20 m s^{-1} overtakes a man running in the same direction in 20 seconds. How long the train will take to overtake the man if he was running in opposite direction?

(1) $\frac{10}{3}$ s
(2) $\frac{20}{3}$ s
(3) $\frac{5}{3}$ s
(4) 3 s

9. Two bodies A (mass 2 kg) and B (mass 3 kg) are dropped from heights of 16 m and 25 m respectively. The ratio of time taken by them to reach ground is

(1) $\frac{3}{5}$
(2) $\frac{4}{5}$
(3) $\frac{1}{5}$
(4) $\frac{3}{4}$

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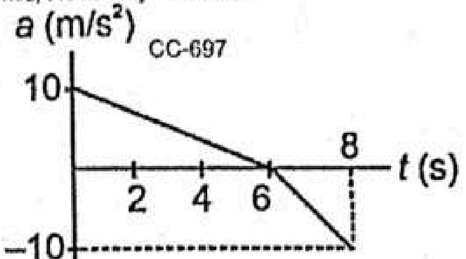
10. Consider the following statements:
Statement (A): If displacement of a particle is zero, the distance covered must be zero.
Statement (B): The numerical value of the ratio of magnitude of average velocity to average speed is always equal to one.
The correct statement(s) is/are
(1) Only statement (A)
(2) Only statement (B)
(3) Both (A) and (B)
(4) Neither (A) nor (B)

11. The variation of velocity (in m s^{-1}) of an object with time (in second) is given by
$v(t) = 6 - 5t + t^2$
The object is at rest at
(1) $t = 2$ s
(2) $t = 3$ s
(3) $t = 6$ s
(4) Both (1) and (2)

12. An object travels 40 m in 5 s and then another 40 m in 10 s. Average speed of the object for complete journey is
(1) 11.44 m/s
(2) 5.33 m/s
(3) 7 m/s
(4) 2.45 m/s

13. The acceleration-time graph for a particle moving along x-axis is shown in figure. If the initial velocity of particle is –5 m/s, the velocity at t = 8 s is



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(1) +15 m/s
(2) +20 m/s
(3) –15 m/s
(4) –20 m/s

14. A particle is moving along x-axis such that velocity is given as $v = (x^2 + x + 3)$ m/s, where x is in m. The acceleration of the particle at x = 1.0 m is
(1) 5 m/s^2
(2) 15 m/s^2
(3) 3 m/s^2
(4) $\frac{5}{3}$ m/s^2

15. If a_r and a_t represents the magnitude of radial and tangential acceleration, the motion of the particle will be uniform circular motion if
(1) $a_r = 0, a_t = 0$
(2) $a_r = 0, a_t \neq 0$
(3) $a_r \neq 0, a_t \neq 0$
(4) $a_r \neq 0, a_t = 0$

16. The equation of trajectory of an oblique projectile is $y = \sqrt{3}x - \frac{gx^2}{2}$. The angle of projection is
(1) 30°
(2) 45°
(3) 60°
(4) 15°

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17. At any instant a motor car is travelling at 60 m/s on a circular road of radius 1200 m. It is increasing its speed at the rate of 4 m/s^2. The magnitude of acceleration of the car is
(1) 3 m/s^2
(2) 5 m/s^2
(3) 7 m/s^2
(4) 4 m/s^2

18. A body is thrown horizontally from a tower, 100 m high, with a velocity 10 m/s. It is moving at an angle 45° with the horizontal after (g = 10 m/s^2)
(1) 2 s
(2) 4 s
(3) 1 s
(4) 3 s

19. A body is projected with a velocity of $(3\hat{i} + 4\hat{j})$ m/s. The maximum height attained by projectile is (g = 10 ms^{-2})
(1) 0.8 m
(2) 8 m
(3) 4 m
(4) 0.4 m

20. The initial position of an object at rest is given by $(3\hat{i} - 8\hat{j})$ m. It moves with constant acceleration and reaches to the position $(2\hat{i} + 4\hat{j})$ m after 4 s. What is its acceleration (in m/s^2)?
(1) $-\frac{1}{8}\hat{i} + \frac{3}{2}\hat{j}$
(2) $2\hat{i} - \frac{1}{8}\hat{j}$
(3) $-\frac{1}{3}\hat{i} + 8\hat{j}$
(4) $8\hat{i} - \frac{3}{2}\hat{j}$

21. Rain is falling vertically with a speed of 35 m s^{-1}. A woman rides a bicycle with a speed of 12 m s^{-1} in east to west direction. In which direction should she hold her umbrella?
(1) $\tan^{-1}\left(\frac{12}{35}\right)$ with vertical
(2) $\tan^{-1}\left(\frac{35}{12}\right)$ with vertical
(3) $\tan^{-1}\left(\frac{12}{35}\right)$ North of West
(4) $\tan^{-1}\left(\frac{35}{12}\right)$ North of West

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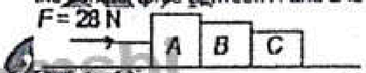
22. Three blocks A, B and C of masses 8 kg, 4 kg and 2 kg respectively, are in contact on a frictionless surface, as shown. If a force of 28 N is applied on the 8 kg block, then the contact force between A and B is

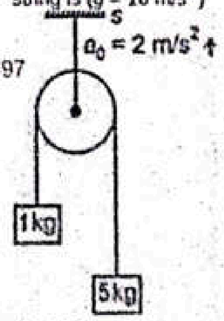


(1) 12 N
(2) 16 N
(3) 36 N
(4) 4 N

23. A bend in a level road has a radius of curvature 100 m. The maximum speed with which a car can negotiate this bend without skidding, will nearly be equal to (μ = 0.8)
(1) 20 m/s
(2) 15 m/s
(3) 28 m/s
(4) 40 m/s

24. A coin placed on a rotating turntable just slips if it is placed at a distance of 4 cm from the centre. If the angular velocity of turntable is doubled, it will just slip at a distance of
(1) 1 cm
(2) 2 cm
(3) 3 cm
(4) 4 cm

25. In the figure shown two blocks of masses 1 kg and 5 kg are attached to a string over a massless pulley. Support moves with an upward acceleration of 2 m/s^2, the tension in the string is (g = 10 m/s^2)



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(1) 10 N
(2) 20 N
(3) 30 N
(4) 50 N

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26. Given below are two statements one is assertion (A) the other is reason (R).

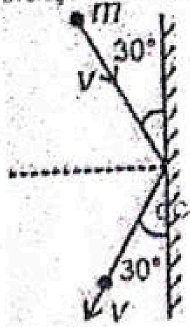
Assertion (A) : Kinetic energy with any reference is always positive.

Reason (R) : Kinetic energy is frame dependent.

In the light of above statements, choose the **correct answer.**

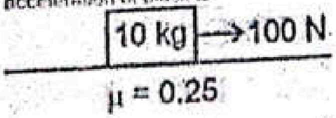
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

27. A ball of mass 0.5 kg moving with speed $v = 20\ m\ s^{-1}$ strikes a hard wall at an angle of 30° with the wall as shown. If the ball is in contact with the wall for 0.05 seconds, then average force acting on the ball is



(1) 173.2 N
(2) 346.2 N
(3) 200 N
(4) 100 N

28. A block of mass 10 kg is under the action of horizontal force 100 N, placed on a rough horizontal surface (μ = 0.25). The acceleration of block is



(1) $10\ m/s^2$
(2) $2.5\ m/s^2$
(3) $7.5\ m/s^2$
(4) Zero

29. A ball collides with a smooth horizontal surface in xz-plane with a velocity $\vec{v} = 2\hat{i} - 3\hat{j}$. If the coefficient of restitution is $\frac{1}{3}$, then the velocity of the ball just after the impact will be

(1) $-\frac{2}{3}\hat{i} - 3\hat{j}$
(2) $\frac{1}{3}\left(2\hat{i} + 3\hat{j}\right)$
(3) $2\hat{i} + \hat{j}$
(4) $\frac{2}{3}\hat{i} - \hat{j}$

30. A block of mass 4 kg accelerates uniformly from rest to a speed of 5 m/s in 20 s. The average power developed in time interval of 0 to 20 s is

(1) 1 W
(2) 3 W
(3) 2.5 W
(4) 5 W

31. Two springs, S_1, and S_2 have spring constants 200 N/m and 400 N/m respectively. For same spring force in the springs, the energy stored in the springs are in ratio

(1) 4 : 3
(2) 1 : 6
(3) 2 : 1
(4) 8 : 1

32. If $F = 2x^2 - 3x - 2$, then select the correct statement

(1) $x = -\frac{1}{2}$ is the position of stable equilibrium
(2) $x = 2$ is the position of stable equilibrium
(3) $x = -\frac{1}{2}$ is the position of unstable equilibrium
(4) $x = 2$ is the position of neutral equilibrium

33. A pendulum of length 'l' has a bob of mass 'm'. If it is released at an instant when string makes an angle of 53° with the vertical, the tension in the string at the lowest position is

(1) $\frac{7mg}{5}$
(2) $\frac{9mg}{5}$
(3) $\frac{3mg}{5}$
(4) $\frac{11mg}{5}$

34. A spring of force constant 500 N/m has an extension of 4 cm. The work done in extending it from 4 cm to 12 cm is

(1) 3.2 J
(2) 1.6 J
(3) 0.8 J
(4) 6.4 J

35. A force $F = \left(3x^2 + 2x\right)$ N displaces a body of mass 1 kg from $x = 0$ to $x = 3$ m. Work done by the force in displacing the body is

(1) 36 J
(2) 81 J
(3) 99 J
(4) 108 J

36. When a conservative force does negative work on a body, the potential energy of the body

(1) Increases
(2) Decreases
(3) First increases and then decreases
(4) Remains unaltered

37. The moment of inertia of a ring of radius R and mass m about an axis passing from the edge of ring and normal to the plane of ring is

(1) mR^2
(2) $2mR^2$
(3) [illegible]
(4) $3mR^2$

38. Two parallel forces 20 N and 30 N act at the ends A and B of a rod 1.5 m long. If an equivalent force is applied instead at a point on the rod in order to produce the same torque, then it must act at

(1) 90 cm from A
(2) 75 cm from B
(3) 20 cm from B
(4) 85 cm from A

39. Two bodies of mass 10 kg and 2 kg are moving with velocities $2\hat{i} - 7\hat{j} + 3\hat{k}$ and $-10\hat{i} + 35\hat{j} - 3\hat{k}$ m/s respectively. The velocity of their centre of mass is

(1) $2\hat{i}$ m/s
(2) $3\hat{k}$ m/s
(3) $\left(2\hat{j} + 2\hat{k}\right)$ m/s
(4) $2\left(\hat{i} + \hat{j} + \hat{k}\right)$ m/s

40. A 500 kg boat is 9 m long and is floating without motion on still water. A man of mass 100 kg is at one end and move to other end of boat and stops. The displacement of boat is

(1) 1.5 m in the direction of displacement of man
(2) 0.75 m in the direction of displacement of man
(3) 1.5 m in the direction opposite to displacement of man
(4) 0.75 m in the direction opposite to displacement of man

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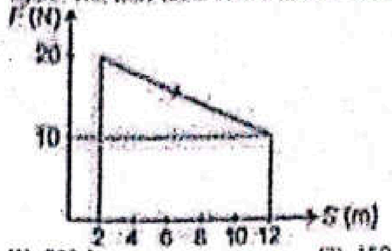
41. Consider the following statements:
(A) In pure translational motion at any instant of time, all particles of the body have the same acceleration.
(B) In pure rotational motion of a rigid body about a fixed axis, every particle of the body (except fixed point) has same angular velocity.
(C) The motion of a rigid body which is not pivoted is always a combination of translation and rotation.
The correct statement(s) is/are
(1) Only (A)
(2) Only (B)
(3) Both (A) and (B)
(4) Both (A) and (C)

42. A wheel of moment of inertia 5×10^{-3} kg m^2 is making 20 revolutions per second. The magnitude of torque required to uniformly stop it in 10 s, is
(1) $4\pi \times 10^{-2}$ N m
(2) $2\pi \times 10^{-2}$ N m
(3) $6\pi \times 10^{-2}$ N m
(4) $8\pi \times 10^{-2}$ N m

43. If the angular momentum of a rotating body about a fixed axis is increased by 20%, keeping its moment of inertia constant, then its kinetic energy will be increased by
(1) 54%
(2) 60%
(3) 30%
(4) 69%

44. The net external force on a body is zero, then according to Newton's second law
(1) The body must remain at rest
(2) The body must have non-zero acceleration
(3) The body may move with uniform velocity or at rest
(4) The momentum of the body must be zero

45. The force-displacement graph of a body is as shown in figure. The work done from 2 m to 12 m is



(1) 200 J
(2) 150 J
(3) 100 J
(4) 175 J

CHEMISTRY

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46. Mass percentage of sulphur in SO_3 is (Atomic mass of sulphur = 32 u)
(1) 25%
(2) 60%
(3) 50%
(4) 40%

47. From 220 mg of CO_2, 10^{21} molecules of CO_2 are removed. Number of remaining molecules of CO_2 will be
(1) 3.011×10^{21}
(2) 2.011×10^{21}
(3) 1.011×10^{21}
(4) 6.022×10^{21}

48. The largest number of molecules are present in
(1) 4 g H_2
(2) 28 g N_2
(3) 0.5 mole O_2
(4) 11 g CO_2

49. An organic compound containing C, H and N gave the following analysis C = 40%; H = 13.33%; N = 46.67%. Its empirical formula would be
(1) CH_4N
(2) CH_5N
(3) $C_2H_7N_2$
(4) CH_3N

50. Match the column I with column II.

	Column I (Prefix)		Column II (multiple)
a.	mega	(i)	10^{15}
b.	giga	(ii)	10^{12}
c.	tera	(iii)	10^{9}
d.	peta	(iv)	10^{6}

Choose the most appropriate match from the option given below:
(1) a(iv), b(iii), c(ii), d(i)
(2) a(iv), b(i), c(ii), d(iii)
(3) a(iii), b(i), c(ii), d(iv)
(4) a(iii), b(ii), c(iv), d(i)

CC-697

CC-697

51. When 1 dm^3 of CO_2 gas is passed over hot coke the volume of gaseous mixture after complete reaction at STP becomes 1.4 dm^3. The composition of the gaseous mixture at STP is
(1) [illegible] of CO, 0.8 dm^3 of CO_2
(2) [illegible] of CO, 0.6 dm^3 of CO_2
(3) [illegible] of CO, 0.8 dm^3 of CO_2
(4) [illegible] dm^3 of CO, 0.4 dm^3 of CO_2

52. [illegible] fraction of solvent in aq. H_2SO_4 solution is 0.85 then number of moles of solute present in 2 kg of solvent is
(1) 6.66
(2) 5.66
(3) 0.15
(4) 19.6

53. Which of the following pairs of compounds follow law of multiple proportions?
(1) H_2O and Na_2O
(2) CO_2 and CO
(3) CO_2 and NO_2
(4) BaO and BeO

54. Consider the following statements
(a) 1 amu = 1.66056×10^{-27} kg
(b) 1 mole of atomic H contains 6.022×10^{-23} hydrogen atoms
(c) SI unit of luminous intensity is candela
The correct statements are
(1) (a) and (b) only
(2) (b) and (c) only
(3) (a) and (c) only
(4) (a), (b) and (c)

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55. If the density of ethanol sample is 0.8 [illegible] then the volume of ethanol needed to make 2L of 0.2 M aqueous solution is
(1) 10 mL
(2) 18 mL
(3) 23 mL
(4) 28 mL

56. Neutral oxide among the following is
(1) Na_2O
(2) CO
(3) As_2O_3
(4) CO_2

57. Identify the **incorrect** statement from the following :

(1) The largest and the smallest species among Mg, Mg^{2+}, Al and Al^{3+} are Al and Mg^{2+} respectively.

(2) The IUPAC name of the element with atomic number 107 is Unnilseptium.

(3) The similarity in behaviour of Li with Mg is referred to as 'diagonal relationship'.

(4) The oxidation state and covalency of Al in $[AlCl(H_2O)_5]^{2+}$ are 3 and 6, respectively.

58. Correct order of ionisation energy of Li, Be, B and C is

(1) C > Be > B > Li
(2) Be > C > B > Li
(3) C > B > Be > Li
(4) B > Be > C > Li

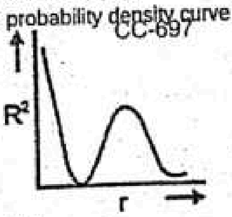
59. The correct order of increasing metallic character of Na, Be, P, Mg and Si is

(1) P < Si < Be < Mg < Na
(2) P < Si < Na < Mg < Be
(3) P < Mg < Be < Si < Na
(4) Be < Si < P < Mg < Na

60. What is the correct order of negative value of electron gain enthalpy of group 17 elements?

(1) Cl > F > Br > I
(2) F > Cl > Br > I
(3) Cl > F > I > Br
(4) F > Br > Cl > I

61. The probability density curve for 2s electron appears like

(1)


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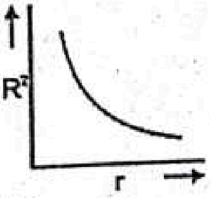

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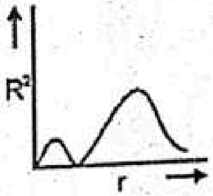

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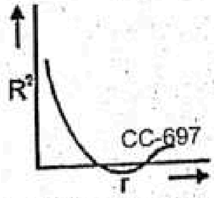

62. The orbital angular momentum of electron in 4s orbital of Cu atom is

(1) $\sqrt{2}\,\hbar$
(2) $2\sqrt{3}\,\hbar$
(3) Zero
(4) $\sqrt{6}\,\hbar$

63. Match the following items in column I with column II and choose the correct option.

	Column I (quantum nos.)		Column II (orbital)
a.	n = 2, l = 1	(i)	3*p*
b.	n = 3, l = 2	(ii)	3*d*
c.	n = 1, l = 0	(iii)	2*p*
d.	n = 3, l = 1	(iv)	1*s*

(1) a(iii), b(i), c(iv), d(ii)
(2) a(iii), b(ii), c(iv), d(i)
(3) a(ii), b(i), c(iii), d(iv)
(4) a(i), b(ii), c(iii), d(iv)

64. Given below are two statements.

Statement I: Electromagnetic waves do not require medium and can move in vacuum.

Statement II: The wavelength of radio waves is less than that of microwave.

In the light of above statements, choose the **correct** option.

(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

65. A bulb is rated at 150 watt, converting 8% energy into light. If energy of one photon is 4.42×10^{-19} J, how many photons are emitted by the bulb per second?

(1) 2.71×10^{19}
(2) 4.06×10^{19}
(3) 27.2×10^{19}
(4) 1.35×10^{19}

66. If the work function of metal is 3.4 eV and the kinetic energy of ejected electron is 1.6 eV then the energy of radiation falling on the metal surface is

(1) 1.8 eV
(2) 4.2 eV
(3) 5.0 eV
(4) 6.2 eV

67. A 3.3 mg ball is moving with a speed of 200 m/s. The de Broglie wavelength associated with it will be ($h = 6.6 \times 10^{-34}$ Js)

(1) 10^{-27} m
(2) 10^{-30} m
(3) 10^{-24} m
(4) 10^{-33} m

68. The number of lines obtained in UV region of hydrogen spectrum when electron jumps from 6th excited state to 1st excited state is/are:

(1) 5
(2) Zero
(3) 4
(4) 3

69. If the kinetic energy of a particle is tripled, its de-Broglie's wavelength becomes

(1) 2 times
(2) 4 times
(3) $\sqrt{2}$ times
(4) $\frac{1}{\sqrt{3}}$ times

70. Select incorrect statement(s) from the following.

(i) For n = 3, total number of orbitals is 9.

(ii) Two electrons occupying the same orbital are distinguished by magnetic orbital quantum number.

(iii) There are (n – 1) subshells in the nth shell.

Choose the correct option.

(1) (i) only
(2) (iii) only
(3) (i) and (ii) only
(4) (ii) and (iii) only

71. The hybridisation of Xe in XeF_6 is

(1) sp^3d
(2) dsp^2
(3) sp^3d^3
(4) sp^3d^2

72. An example of odd-electron molecule is

(1) F_2
(2) BCl_3
(3) NO_2
(4) O_3

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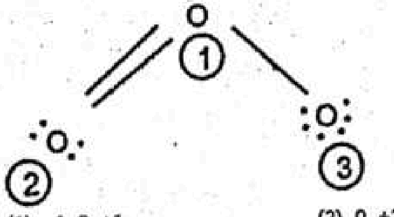
73. Which of the following is diamagnetic?
(1) NO (2) N_2
(3) O_2^- (4) N_2^+

74. The correct formal charges on oxygen atoms numbered 2, 1 and 3 respectively are :



(1) –1, 0, +1 (2) 0, +1, –1
(3) 0, 0, 0 (4) +1, 0, –1

75. Identify the correct statement about ClF_3 from the following options :
(1) It has T-shaped geometry with two lone pairs on Cl atom.
(2) It has T-shaped geometry with three lone pairs on Cl atom.
(3) It has a trigonal pyramidal geometry with two lone pairs on Cl atom.
(4) It has a planar trigonal geometry with two lone pairs on Cl atom.

76. Given below are two statements, one is labelled as **Assertion (A)** and other is labelled as **Reason (R)**.
Assertion (A): The resultant dipole moment of NH_3 is greater than that of NF_3.
Reason (R): In case of NH_3, the orbital dipole due to lone pair is in the same direction as the resultant dipole moment of N-H bonds whereas orbital dipole is in opposite direction to that of N-F bonds in NF_3.
In the light of above statements, choose the **correct** option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

77. The hybridization of atomic orbitals of xenon in XeO_3, XeF_4 and XeF_2 respectively are
(1) sp^2, sp^3d^2 and sp^3d (2) sp^3, sp^3d and sp^3d^2
(3) sp^3, sp^3 and sp (4) sp^3, sp^3d^2 and sp^3d

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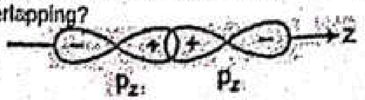
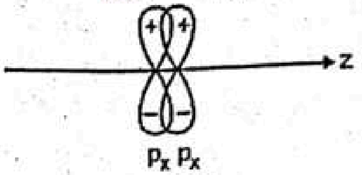
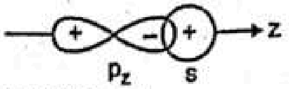
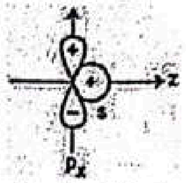
78. Which of the following diagram represents negative overlapping?

(1) 

(2) 

(3) 

(4) 

79. Consider the following statements about molecular orbital theory.
(a) Molecular orbital is monocentric.
(b) Number of molecular orbitals formed is equal to the number of combining atomic orbitals.
(c) The bonding molecular orbital has lower energy and hence greater stability than the corresponding antibonding molecular orbital.
The correct statements are
(1) a and b only (2) b and c only
(3) a and c only (4) a, b and c

80. The correct order of bond length of C – C, C – N, C – H is
(1) C – H < C – N < C – C (2) C – C < C – N < C – H
(3) C – C < C – H < C – N (4) C – N < C – H < C – C

81. The enthalpies of combustion of carbon and carbon monoxide are –390 kJ and –278 kJ respectively then the enthalpy of formation of CO in kJ is
(1) –668 (2) 668
(3) 112 (4) –112

82. At a certain temperature, T (K), during a process, 500 J is absorbed by the system and work of 200 J is done by the system. Then change in internal energy of the system is :
(1) 400 J (2) 300 J
(3) 700 J (4) 500 J

83. One mole of an ideal gas undergoes change of state from (4 atm, 2L, 100 K) to (6 atm, 3L, 310 K) with a change in internal energy $\Delta U = 34$ L atm. The change in enthalpy (ΔH) of the process is
(1) 44 L atm
(2) 34 L atm
(3) 24 L atm
(4) Not defined as pressure is not constant

84. In which of the following processes entropy decreases?
(a) Boiling of egg.
(b) Temperature of a crystalline solid lowered from 270K to 170K.
(c) $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$
(d) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$
Choose the correct answer from options give below.
(1) (a) and (c) only (2) (a), (b) and (d) only
(3) (b) and (d) only (4) (b), (c) and (d) only

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85. The standard enthalpy of formation of $H_2O(l)$ and glucose (s) at 25°C are -300 kJ mol^{-1} and -1300 kJ mol^{-1} respectively and standard enthalpy of combustion of C (graphite) is -400 kJ mol^{-1}. The ΔU of combustion of glucose per gram at 25°C will be
(1) -3004 kJ
(2) -2900 kJ
(3) -8.48 kJ
(4) -16.11 kJ

86. Which of the following is correct option for free expansion of an ideal gas under adiabatic condition?
(1) $q = 0, \Delta T < 0, w \neq 0$
(2) $q = 0, \Delta T \neq 0, w = 0$
(3) $q \neq 0, \Delta T = 0, w = 0$
(4) $q = 0, \Delta T = 0, w = 0$

87. 4 L of an ideal gas at a pressure of 20 atm expands isothermally at 25°C into a vacuum until its total volume became 20 L. Work done in the expansion is
(1) $w = -320$ atm L
(2) $w = 320$ atm L
(3) $w = 32$ atmL
(4) $w = 0$

88. Which among the following is an intensive property?
(1) Internal energy
(2) Enthalpy
(3) Heat capacity
(4) Refractive index

89. The suitable thermodynamic conditions for a spontaneous reaction at all temperature are
(1) $\Delta H > 0, \Delta S < 0$
(2) $\Delta H < 0, \Delta S > 0$
(3) $\Delta H > 0, \Delta S > 0$
(4) $\Delta H < 0, \Delta S < 0$

90. In which of the following reaction $\Delta H = \Delta U$
(1) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
(2) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
(3) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$
(4) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

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91. Which structure determines the shape of the cell and provides a strong structural support to prevent the bacterium from bursting?
(1) Slime layer
(2) Capsule
(3) Cell membrane
(4) Cell wall

92. The major function of Golgi apparatus is
(1) Packaging of materials
(2) To synthesise steroidal hormones
(3) Providing sites for protein synthesis
(4) Storage of fats

93. Match the following column I with column II and select the correct option.

	Column I		Column II
a.	Stroma lamellae	(i)	A space limited by the inner membrane of the chloroplast
b.	Stroma	(ii)	Stacked structures found within the chloroplast of plant cells
c.	Thylakoids	(iii)	Flat membranous tubules that connect the organized flattened membranous sacs of different grana
d.	Grana	(iv)	Flattened membranous sacs present in matrix of chloroplast

(1) a(ii), b(i), c(iv), d(iii)
(2) a(iii), b(i), c(iv), d(ii)
(3) a(iii), b(iv), c(i), d(ii)
(4) a(ii), b(iv), c(i), d(iii)

94. Centrioles differ from axoneme of cilia or flagella in
(a) Total number of microtubules present at periphery.
(b) Presence of radial spokes.
(c) Arrangement of microtubules.
(d) Containing proteinaceous central hub.
(e) Being surrounded by double membrane.
The correct ones are
(1) (a), (c) and (d) only
(2) (a), (b), (c) and (d)
(3) (a), (b) and (c) only
(4) (b), (c), (d) and (e)

95. The A of a polysome translate the B into proteins.
Select the correct option for A and B.
(1) A – Ribosomes, B – mRNA
(2) A – mRNA, B - Ribosome
(3) A – ER, B - DNA
(4) A – DNA, B - ER

96. Choose the **incorrect** statement about the main arena of cellular activities of a cell.
(1) It is a semi-fluid matrix
(2) It is found in eukaryotes only
(3) It occupies the volume of cell
(4) It contains enzymes and cell organelles

CC-697

97. Read the following statements and select the **correct** option.
Statement A: Chromatin in eukaryotes contains DNA and some basic proteins called histones.
Statement B: The cell wall and middle lamellae may be traversed by plasmodesmata which connect the cytoplasm of neighbouring plant cells.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are incorrect
(4) Both statements A and B are correct

98. Synthesis of steroidal hormones occurs in
(1) SER
(2) RER
(3) Golgi apparatus
(4) Mitochondria

99. Inclusion bodies
(1) Are found only in eukaryotes
(2) Are the forms of stored reserve material
(3) Are single membrane bound
(4) Include vacuole with a tonoplast

100. Satellite in chromosomes is
(1) Seen in telocentric chromosomes only
(2) A fragment beyond the secondary constriction
(3) Formed by primary constriction
(4) Devoid of DNA

101. Ribosomes are composed of
(1) rRNA and tRNA
(2) mRNA and proteins
(3) mRNA and rRNA
(4) rRNA and proteins

102. At the end of anaphase II, how many chromatids are present in each chromosome?
(1) One
(2) Two
(3) Four
(4) Eight

103. Liquid endosperm in coconut exemplifies a structure that is formed when
(1) Karyokinesis is not followed by cytokinesis
(2) Cytokinesis is achieved by cell plate formation
(3) Karyokinesis is delayed
(4) Four haploid uninucleate progenies are formed

104. Telophase is characterised by
- (1) Splitting of centromere
- (2) Reappearance of nuclear membrane
- (3) Shortening of the spindle fibres
- (4) Condensation of chromosomes into chromatin fibres

105. In plant cells cytokinesis is achieved by formation of
- (1) Cell furrow
- (2) Cell membrane
- (3) Cell plate
- (4) Centrioles

106. The phase of the cell cycle where cells remain metabolically active but usually do not proliferate is
- (1) G_0 phase
- (2) M-phase
- (3) S-phase
- (4) G_1 phase

107. The stage which lasts for months or years in the oocyte of some vertebrates is
- (1) Zygotene
- (2) Leptotene
- (3) Diplotene
- (4) Pachytene

108. In the S phase of cell cycle
- (1) Chromosome number increases
- (2) Cells become haploid
- (3) Amount of DNA doubles
- (4) Amount of DNA becomes half

109. Metaphase I is different from metaphase II as in former
- (1) Bivalents align at equator
- (2) Each chromosome is attached to the spindle fibres from both the poles
- (3) Single equatorial plate is formed
- (4) Each chromosome has single chromatid

110. What will be the content of DNA in a somatic cell at G_2 if its meiotic product has 20 picogram of DNA?
- (1) 40 pg
- (2) 20 pg
- (3) 80 pg
- (4) 160 pg

111. Select the mismatched pair.
- (1) Class – Mammalia
- (2) Order – Carnivora
- (3) Family – Felidae
- (4) Genus – Convolvulaceae

112. In which of the following taxonomic categories organisms belong to the same taxon have the maximum common characters?
- (1) Genus
- (2) Order
- (3) Class
- (4) Division

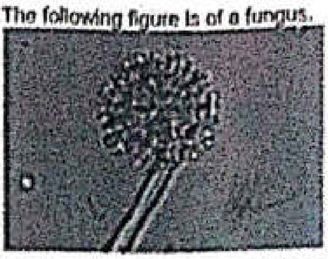
113. Select the **incorrect** statement for rules of Binomial Nomenclature
- (1) Each name has two components – the generic name and the specific epithet
- (2) Biological names are generally in Latin and printed in italics
- (3) The names are Latinised or derived from Latin irrespective of their origin
- (4) Name of the author appears before specific epithet i.e. at the end of generic name and is written in an abbreviated form

114. Cat and lion belong to
- (1) Two different genera but the same species
- (2) Two different genera but the same family
- (3) The same order but different families
- (4) Two different orders and the same family

115. Who is known as 'The Darwin of the 20th century'?
- (1) Ramdeo Misra
- (2) Birbal Sahni
- (3) Ernst Mayr
- (4) Carolus Linnaeus

116. Modern taxonomic studies includes all of the following, **except**
- (1) External features
- (2) Structure of cell
- (3) Developmental process
- (4) Evolutionary studies

117. The following figure is of a fungus.



Which of the given statements is not true for this fungus?
- (1) It is commonly known as a sac-fungus
- (2) It produces conidia endogenously
- (3) Its mycelium is branched and septate
- (4) It forms fruiting body called ascocarp

118. Which of the following is not true w.r.t. cyanobacteria?
- (1) They are eubacteria
- (2) They are majorly decomposers
- (3) They have rigid cell wall
- (4) They are also referred to as blue green algae

119. Select the incorrect w.r.t diatoms.
- (1) Cell membrane is impregnated with silica
- (2) They form diatomaceous earth
- (3) They are known as chief producers of oceans
- (4) They are included in Chrysophytes

120. Phycobiont of lichens
- (1) Absorbs minerals from the soil
- (2) Is heterotrophic in nature
- (3) Prepares food for its partner component
- (4) Absorbs water from the humid atmosphere

121. Choose the incorrect statement w.r.t. slime moulds.
- (1) Under suitable conditions they form an aggregation called plasmodium
- (2) During unfavourable conditions plasmodium differentiates and forms fruiting bodies
- (3) The spores covered by protein rich layer known as pellicle
- (4) The spores are dispersed by air currents

122. The spherical bacteria are
- (1) Bacilli
- (2) Cocci
- (3) Vibrio
- (4) Spirilla

123. The parasitic fungi causing white rust on mustard is
- (1) *Rhizopus*
- (2) *Albugo*
- (3) *Alternaria*
- (4) *Claviceps*

124. Bacteriophages usually have
- (1) dsDNA
- (2) ssDNA
- (3) dsRNA
- (4) ssRNA

125. Sleeping sickness causing protozoan is parasitic in nature. It
- (1) Is ciliated
- (2) Has pseudopodia
- (3) Is flagellated
- (4) Lacks locomotory structure

126. There may be variation in length of filaments of stamens within a flower, as in
- (1) Pea
- (2) China rose
- (3) *Salvia*
- (4) *Petunia*

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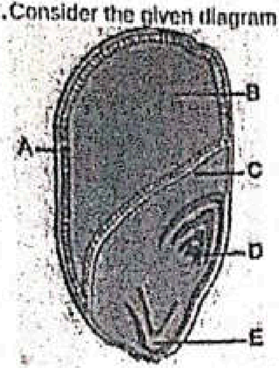
127. Consider the given diagram and choose the correct option



(1) Label A represents the lipid rich layer which is the covering of the embryo of the seed
(2) B is the result of double fertilization
(3) Label D and E are enclosed in a sheath known as coleorhiza and coleoptile respectively
(4) Structure labelled as C is scutellum found in both monocot and dicot seeds

128. Select the correct option w.r.t. flower with diadelphous stamen
(1) *Citrus*
(2) China rose
(3) Pea
(4) Lily

CC-697

129. Symbol ⊕, in floral formula shows the flower is
(1) Zygomorphic
(2) Actinomorphic
(3) Bisexual
(4) Unisexual

130. Free central placentation is seen in
(1) Mustard
(2) *Argemone*
(3) Tomato
(4) *Dianthus*

131. In racemose inflorescence
(1) Flowers are borne in basipetal order
(2) Main axis continues to grow
(3) Growth of main axis is limited
(4) Younger flower are present at the base and older flowers are present towards the apex

132. Read the given statements of **assertion (A)** and **reason (R)** and choose the correct option.

Assertion(A): Cooling effect in leaves and bringing fresh air to the leaf surface is said to be facilitated by petiole.

Reason(R): The long, thin, cylindrical, flexible petiole allows leaf blade to flutter in wind.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

133. In neem,
(1) The leaflets are attached at a common point, *i.e.*, at the tip of petiole
(2) Lamina is entire
(3) A number of leaflets are present on a common axis, the rachis, which represents the midrib of the leaf
(4) Lamina is incised and the incisions do not touch the midrib

CC-697

134. Mango and coconut develop from
(1) Monocarpellary superior ovary
(2) Multicarpellary superior ovary
(3) Monocarpellary inferior ovary
(4) Multicarpellary inferior ovary

135. Green leaf like structure that protect the flower in bud stage is
(1) Sepal
(2) Carpel
(3) Androecium
(4) Gynoecium

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136. A loose connective tissue is composed of all of the following, **except**
(1) Matrix, cells and fibres
(2) Mixture of modified polysaccharides
(3) Different types of cells present in matrix
(4) Only one type of fibres

137. Dense connective tissues consist of compactly packed fibres and fibroblasts and it can be further categorised into dense regular and dense irregular tissues on the basis of
(1) Orientation of fibres
(2) Origin of fibres
(3) Number of fibres
(4) Structural composition of fibres

138. The type of epithelium present in the inner lining of ducts of salivary gland, is also present in the
(1) Walls of blood vessels
(2) Lining of stomach and intestine
(3) Inner surface of hollow organs like bronchioles
(4) Moist surface of buccal cavity

139. Match column I with column II and choose the correct option.

	Column I		Column II
a.	Tight junctions	(i)	Facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells
b.	Adhering junctions	(ii)	Present in cardiac muscle tissue
c.	Gap junctions	(iii)	Perform cementing to keep neighbouring cells together
d.	Communication junctions	(iv)	Help to stop substances from leaking across a tissue

CC-697

(1) a(iii); b(iv); c(ii); d(i)
(2) a(iv); b(iii); c(i); d(ii)
(3) a(iii); b(iv); c(i); d(ii)
(4) a(iii); b(ii); c(iv); d(i)

140. Choose the incorrect statement w.r.t. neuroglial cells of the nervous tissue.
(1) These are the cells that provide insulation to the nerve cells
(2) These are non-excitable cells and provide protection to the neurons
(3) They form the majority of the neural tissue
(4) They are the unit of neural system

141. Characteristic of smooth muscle fibres are
(1) Spindle shaped, unbranched, uninucleated, unstriated, voluntary
(2) Spindle shaped, unbranched, uninucleated, unstriated, involuntary
(3) Cylindrical, unbranched, striated, multinucleated, voluntary
(4) Cylindrical, branched, striated, uninucleated, involuntary

142. The matrix of _A_ is solid and pliable but that of _B_ is hard and non-pliable.
Choose the option that **correctly** identifies A and B respectively.
(1) Bone, cartilage
(2) Cartilage, bone
(3) Blood, cartilage
(4) Tendon, ligament

143. Cuboidal epithelium is **not** present in
(1) Fallopian tubes
(2) PCT of nephron
(3) Ducts of glands
(4) DCT of nephron

144. The enormous diversity of protein molecules is mainly due to diversity of
(1) Carboxyl groups on the amino acids
(2) Amino groups on the amino acids
(3) Peptide bonds
(4) Amino acid sequences within the protein molecule

145. The total number of hydroxyl groups in a glycerol molecule is equal to the total number of
(1) Hydroxyl group attached to the heterocyclic ring in adenine
(2) Hydroxyl group attached to the glucose molecule
(3) Nitrogen atoms present in the adenine
(4) Fatty acids present in a triglyceride

146. A diglyceride differs from a monoglyceride in
(1) Having two molecules of trihydroxypropane
(2) Lacking one unsaturated fatty acid
(3) Having one additional molecule of glycerol
(4) Possessing one additional fatty acid

147. Which is not correct about enzyme activity?
(1) There is an obligatory formation of an 'ES' complex in every enzyme catalyzed reaction.
(2) If 'product' is at a lower energy level than the 'substrate' then one need not to supply energy in order to form the product.
(3) If product is at high energy level than the substrate, then the reaction is exothermic in nature.
(4) Enzymes bring down activation energy, making the transition of 'substrate' to 'product' more easy.

148. Which of the following statements regarding protein is **incorrect?**
(1) Peptide bond formation is a condensation reaction
(2) Denaturation always leads to the loss of primary structure only
(3) Secondary structure is mainly stabilized by the hydrogen bonds
(4) Tertiary structure also involves interactions among R–groups (side chains) of amino acids

149. Hydroxymethyl is present as 'R' group in
(1) Glycine
(2) Serine
(3) Tyrosine
(4) Tryptophan

150. Category of biomolecules that forms 10-15% of the total cellular mass of living tissues are
(1) Carbohydrates
(2) Lipids
(3) Nucleic acids
(4) Proteins

151. Suppose the E.C. number of an enzyme is 4.2.1.1. Identify the class to which this enzyme belongs to and choose the correct option w.r.t. it.
(1) Catalyses removal of groups from substrates by mechanisms other than hydrolysis
(2) Catalyses hydrolysis of ester, ether and peptide bonds
(3) Responsible for the linking together of two compounds
(4) Catalyses oxidoreduction between two substrates

152. The total number of nitrogen atoms present in the ring of uracil is equal to the number of H-bonds present between
(1) Guanine and cytosine in DNA
(2) Adenine and thymine in DNA
(3) Trihydroxypropane and choline
(4) Monomeric units of arachidonic acid

153. Which of the following co-factor simultaneously forms coordinate bonds with side chains at the active site and form one or more coordinate bonds with the substrate?
(1) Prosthetic group
(2) Haem
(3) Coenzyme
(4) Metal ion

154. Certain organic compounds form 3% of the total cellular mass of the living tissue. In polymeric form of such organic compounds, monomeric units are joined together with the help of 'P'. 'P' is also present in all of the following, except
(1) Cotton fibre
(2) Deoxyribonucleic acid
(3) RuBisCO
(4) Glycogen

155. **Assertion (A):** Inhibition of succinic dehydrogenase by malonate is an example of competitive inhibition.
Reason (R): Malonate closely resembles the substrate of succinic dehydrogenase in structure.
In the light of the above statements, choose the correct option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

156. In a hypothetical scenario, if 'E' represents enzyme 'S' represents substrate, 'P' represents product, 'ES' represents enzyme-substrate complex and 'EP' represents enzyme-product complex then, which of the following is the correct representation of an enzyme catalysed biochemical reaction?
(1) $E + S \rightarrow EP \rightarrow ES \rightarrow E + P$
(2) $E + P \rightarrow EP \rightarrow ES \rightarrow E + S$
(3) $E + ES \rightarrow EP \rightarrow ES \rightarrow S + P$
(4) $E + S \rightarrow ES \rightarrow EP \rightarrow E + P$

157. Consider the following:
(a) Non-polymeric in nature
(b) Has 16 carbons including the carboxyl carbon
(c) Belongs to the category of saturated fatty acid
The above given features hold true for which of the following molecules?
(1) Arachidonic acid
(2) Glutamic acid
(3) Cytidylic acid
(4) Palmitic acid

158. Select the incorrect option w.r.t vinblastin.
(1) Belongs to the category of drugs
(2) Plays known roles in normal physiological processes of all animals
(3) It is used for the treatment of cancer
(4) It is a secondary metabolite

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CC-697

159. Select the correct option to complete the analogy w.r.t. respiratory structures.
Terrestrial arthropods : Tracheal tubes :: Aquatic arthropods : ______
(1) Moist cuticle
(2) Gills
(3) Lungs
(4) Green glands

160. **Assertion (A):** Any change in the volume of the thoracic cavity, will be reflected in the lung cavity.
Reason (R): We can directly alter the pulmonary volume and any change in the volume of the thoracic cavity will be reflected in the pulmonary cavity.
In the light of the above statements, choose the correct option.
(1) Both (A) and (R) are true but (R) does not explain (A) correctly.
(2) (A) is true, (R) is false.
(3) Both (A) and (R) are false.
(4) Both (A) and (R) are true and (R) explains (A) correctly.

161. Arrange the following lung volumes and capacities (from lower to higher) in an adult man under normal conditions.
A. Tidal volume
B. Residual volume
C. Inspiratory reserve volume
D. Vital capacity
Choose the correct option.
(1) A < C < B < D
(2) A < D < B < C
(3) A < D < C < B
(4) A < B < C < D

162. Body of a person contains 5 L blood. What amount of O_2 can be delivered by 5 L of oxygenated blood to tissues under normal physiological conditions?
(1) 750 mL
(2) 500 mL
(3) 250 mL
(4) 150 mL

163. The conducting part of human respiratory system does not facilitate
(1) Humidification of inhaled air
(2) Temperature adjustment of inhaled air
(3) Clearing of foreign particles from the inhaled air
(4) Exchange of gases between alveolar air and blood

164. Thoracic chamber is ventrally, dorsally and laterally formed, respectively, by
(1) Vertebral column, ribs, and sternum
(2) Ribs, sternum, and diaphragm
(3) Sternum, vertebral column, and ribs
(4) Diaphragm, sternum, and ribs

165. The diffusion membrane helping in exchange of gases in lungs is made up of three major layers. Which of the following is not included in respiratory diffusion membrane?
(1) Thin squamous epithelium of the alveoli
(2) Endothelium of the alveolar capillaries
(3) Cellular basement substance
(4) Acellular layer between alveolar wall and wall of alveolar capillaries

166. Factor not favourable for formation of oxyhaemoglobin is
(1) High pO_2
(2) Low pCO_2
(3) Lesser H^+ concentration
(4) High temperature

167. A chemosensitive area is situated adjacent to the rhythm centre which is highly sensitive to
(1) CO_2 only
(2) H^+ only
(3) CO_2 and H^+
(4) O_2 and H^+ only

168. Which of the following conditions are found in alveoli of lungs?
(1) High pO_2, low pCO_2 and low H^+
(2) High pH, high pCO_2 and high pO_2
(3) High H^+, low pO_2 and high pCO_2
(4) Low pH, high pCO_2 and low pO_2

169. RBCs contain a very high concentration of an enzyme (X) and minute quantities of the same in plasma too. Identify the enzyme (X) which catalyzes the following given reaction in both directions.

$$CO_2 + H_2O \overset{(X)}{\rightleftharpoons} H_2CO_3 \overset{(X)}{\rightleftharpoons} H^+ + HCO_3^-$$

Choose the correct option.
(1) Catalase
(2) Carbonic anhydrase
(3) Oxygenase
(4) Hydrogen peroxidase

170. Neutrophils and monocytes differ from each other in all of the following, **except**
(1) The shape of their nuclei in mature form
(2) Their contribution to total WBC content of blood
(3) Performing phagocytosis of foreign organisms
(4) Their cytoplasmic contents

171. In humans, neural signals through ANS can
(1) Only increase the rate of heart beat
(2) Only decrease the rate of heart beat
(3) Either increase or decrease the rate of heart beat depending upon the neurotransmitter released
(4) Only decrease the speed of conduction of action potentials through cardiac muscle fibres

172. Choose the correct order of flow of blood in systemic circulation in organisms possessing 4 chambered heart.
(1) Right atrium → Left atrium → Lungs → Aorta
(2) Pulmonary artery → Lungs → Pulmonary vein → Right atrium
(3) Right atrium → Right ventricle → Lungs → Systemic veins
(4) Left ventricle → Aorta → Tissue → Vena cava

173. Select the correct statement w.r.t. formed elements of blood.
(1) Basophils secrete histamine, serotonin and heparin and are thus associated with inflammatory reactions.
(2) Lymphocytes constitute about 2-5% of total WBCs.
(3) Neutrophils are agranular phagocytic cells.
(4) Both B and T lymphocytes are responsible for immune responses of the body and both undergo maturation within the bone marrow only.

174. Antibody common to blood plasma of individual with blood group O^{-ve} and individual with blood group B^{+ve} is/are
(1) Anti-B
(2) Anti-A
(3) Both Anti-B and Anti-A
(4) Anti Rh

175. All of the following are true about human heart, **except**
(1) Heart is a mesodermally derived organ
(2) The atrium and ventricle of the same side are separated by atrio-ventricular septum
(3) Right and left ventricles are separated by inter-ventricular septum
(4) Heart is protected by a single walled membranous bag called pericardium

CC-697
CC-697
CC-697
CC-697
CC-697
Aakash Medical|IIT-JEE|Foundations

176. The first heart sound occurs due to

(a) Closure of tricuspid valve
(b) Opening of semilunar valve
(c) Closure of bicuspid valve
(d) Closure of semilunar valves.

Choose the **correct** option.

(1) (a) and (c)
(2) (b) and (c)
(3) (b) only
(4) (a) and (d)

177. In humans, the walls of arteries are thicker than corresponding veins because

(1) Tunica externa of veins is comparatively thick
(2) Tunica media of veins is comparatively thin
(3) Tunica media of arteries is comparatively thin
(4) Tunica intima of arteries is comparatively thick

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178. Select the **incorrect** statement among the following.

(1) During ventricular systole, tricuspid and bicuspid valves get closed
(2) During joint diastole, tricuspid and bicuspid valves open
(3) During ventricular diastole, semilunar valves open
(4) The SAN generates an action potential which stimulates both atria causing atrial systole

179. A reduction in the number of which of the following can lead to clotting disorders that lead to excessive loss of blood from the body?

(1) Platelets
(2) Erythrocytes
(3) Neutrophils
(4) Monocytes

180. Blood from intestine is carried to the liver via ______ before it is returned to systemic circulation. The **correct** option which fills the blank is

(1) Hepatic vein
(2) Hepatic portal vein
(3) Hepatic artery
(4) Inferior mesenteric artery